

Indian Market Surveillance Analysis Assignment

Context

You are provided with real-world datasets from Indian Market Quality Assessment Project. The data originates from collection forms (C1) and laboratory testing forms (T1) and contains logistics data, oil metadata, and extensive chemical test results.

General Instructions

- Use Python standard libraries for data processing and analysis.
- Avoid hard-coded thresholds; design parameter-driven logic.
- Document assumptions and regulatory logic clearly.
- Partial solutions are acceptable if reasoning is sound (and will be evaluated based on the logic).
- Code clarity, structure, and reasoning are as important as correctness.

Assignment Questions

Q1. Data Cleaning & Standardization [Use the data sets from Subsheet name Sheet_C1FORM]

The column Type of Oil [By Sampler] contains inconsistent oil names due to spelling variations, casing differences, and extra spaces.

- Standardize the oil type reported by the sampler.
- Explain the standardization logic.

Q2. Descriptive Analysis [Use the data sets from Subsheet name Sheet_C1FORM]

- Compute oil-wise sample counts.
- Show city-wise distribution for top oil types.
- Export results to Excel.

Q3. Pass / Fail Determination Using Complex Test Criteria

The T1 dataset contains multiple chemical, physical, and adulteration test parameters. A sample is considered FAIL if any critical parameter violates acceptable limits.

- Design a rule-based system to derive Pass / Fail status for each sample in T1.
[Evaluation Criteria shared in excel]
- Ensure thresholds are configurable (not hard-coded).
- Add a Pass / Fail column and a Failure Reason column.
- Explain how the logic adapts to regulatory changes.

Q4. Sampler vs Lab Mismatch Detection [Use the data sets from Subsheet name Sheet_T1FORM]

- Detect oil type mismatches between sampler and lab.
- Classify minor vs major mismatches.

- Summarize mismatches by oil type and city.

Q5. C1 ↔ T1 Interconnection & Key Insight [Use both the sub-sheets]

You are provided with two datasets: C1 (Collection & Logistics Form) and T1 (Testing & Lab Results Form).

- Identify correct join keys between C1 and T1.
- Build a master dataset capturing the full sample journey.
- Analyze delays and operational bottlenecks.
- Derive at least one key insight only possible via C1–T1 linkage [Operational and Analytical].

Submission Guidelines

- Python scripts / notebooks
- Final master dataset with Pass/Fail status
- Generated outputs (Excel / charts / folders)
- README explaining approach, assumptions, and limitations